

I cad 2023

Ex 1

$$V_{CC} - I_{C1} \left(1 + \frac{1}{\beta_{F1}}\right) \cdot R_{E1} + V_{BE1} - I_{B1} R_{B1} = 0$$

$$I_{C1} = \frac{V_{CC} + V_{BE1}}{\frac{R_{E1} \left(1 + \frac{1}{\beta_{F1}}\right)}{\beta_{F1}} + \frac{R_{B1}}{\beta_{F1}}} = 0.746 \text{ mA}$$

$$V_{CC} - I_{C2} R_{B2} - V_{BE2} - R_{C1} \left(I_{C1} + I_{C2} \left(1 + \frac{1}{\beta_{F2}}\right) \right) = 0$$

$$I_{C2} = \frac{V_{CC} - V_{BE2} - R_{C1} I_{C1}}{\frac{R_{B2}}{\beta_{F2}} + \frac{R_{C1} \left(1 + \frac{1}{\beta_{F2}}\right)}{\beta_{F2}}} = 1 \text{ mA}$$

$$V_{CE1} = R_{C1} \left(I_{C1} + I_{C2} \left(1 + \frac{1}{\beta_{F2}}\right) \right) - \left[V_{CC} - R_{E1} \cdot I_{C1} \left(1 + \frac{1}{\beta_{F1}}\right) \right]$$

$$= -7.9 \text{ V}$$

$$V_{CE2} = V_{CC} - R_{C2} I_{C2} - R_{E2} \left[I_{C1} + I_{C2} \left(1 + \frac{1}{\beta_{F2}}\right) \right]$$

$$= 3.55 \text{ V}$$

$$(I_{C1}, V_{CE1}) = (0.746 \text{ mA}, -7.9 \text{ V})$$

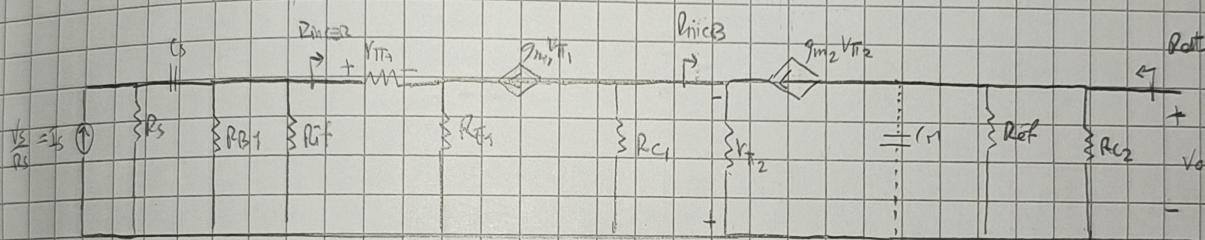
$$(I_{C2}, V_{CE2}) = (1 \text{ mA}, 3.55 \text{ V})$$

small signal analysis: C_s, C_F, C_{B2} shorted, C_{B1} opened
 amplifier topology: current mixing - voltage sampling (shunt)
 Norton equivalent

$$R_{if} = R_F$$

$$R_{of} = R_F$$

$$\beta = \frac{i_i}{V_o / A_{i=0}} = -\frac{1}{R_F}$$



$$A_{OL} = \frac{V_o}{i_s \beta = 0} = \alpha_i \cdot \alpha_{CE1} \cdot \alpha_{CE2}$$

$$A_{OL} = -7.037 [\text{K}\Omega]$$

$$g_{m1} = \frac{I_{C1}}{V_T} = 0.0282 \text{ A/V}$$

$$\alpha_{CE2} = g_{m2} (R_{of} \parallel R_{C2}) = 2149.1 [\text{V/V}]$$

$$A_F = \frac{A_{OL}}{1 + \beta A_{OL}} = -5.332 [\text{K}\Omega]$$

$$r_{\pi1} = \frac{\beta_{01}}{g_{m1}} = 5.23 \text{ K}\Omega$$

$$R_{inCB} = \frac{r_{\pi2}}{1 + \beta_{F2}} = 25.87 \Omega$$

$$A_V = \frac{V_o}{V_s} = \frac{A_F}{R_s} = -5.33 [\text{V/V}]$$

$$g_{m2} = \frac{I_{C2}}{V_T} = 0.0385 \text{ A/V}$$

$$R_{inCE1} = r_{\pi1} + (1 + \beta_{01}) R_{E1} = 76.2 \text{ K}\Omega$$

$$r_{\pi2} = \frac{\beta_{02}}{g_{m2}} = 5.2 \text{ K}\Omega$$

$$\alpha_{CE1} = -\frac{\beta_{01} R_{C1} \parallel R_{inCB}}{R_{inCE1}} = -0.05 [\text{V/V}]$$

$$R_{in}^a = \alpha_i = 944 \Omega$$

$$R_{out}^a = R_{of} \parallel R_{C2} = 3.873 \text{ K}\Omega$$

$$R_{in}^F = \frac{R_{in}^a}{1 + \beta A_{OL}} = 715 \Omega$$

$$R_{out}^F = \frac{R_{out}^a}{1 + \beta A_{OL}} = 2.934 \text{ K}\Omega$$

$$\alpha_i = R_s \parallel R_{B1} \parallel R_{if} \parallel R_{inCE1} = 944 \Omega$$

$$R_{in} = \frac{1}{\frac{1}{R_{in}^F} + \frac{1}{R_s}} = 3.510 \text{ K}\Omega$$

$$R_{out} = R_{out}^F = 2.934 \text{ K}\Omega$$

$$f_H = \frac{1}{2\pi} \frac{1}{C_H R_{HOC}} =$$

$R_{HOC} = R_{of} \parallel R_{C2} \rightarrow$ feedback reduces the resistance seen from $C_H \rightarrow f_H$ is increased, bandwidth enlargement we can apply feedback at f_H

$$f_H^2 = \frac{1}{2\pi} \frac{1}{C_H R_{HOC}} = 4.1 \text{ MHz}$$

$$f_H^4 = (1 + \beta A_{OL}) f_H = 5.41 \text{ MHz}$$

Ex 2

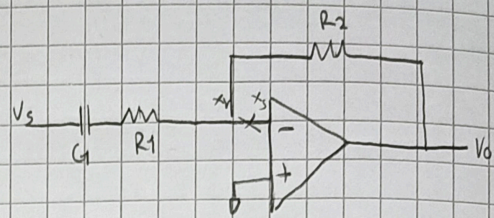
$$R_2 = 22 \text{ k}\Omega$$

$$C_1 = 100 \text{ nF}$$

$$A_{d0} = 10^5 \text{ (V/V)}$$

$$\omega_0 = 10^4 \text{ rad/s}$$

$$A_d(s) = \frac{A_{d0}}{1 + s/\omega_0}$$



with $R_1 = 0.2 \text{ }\Omega$ 1) ideal gain $A_v = \frac{V_o}{V_s} = -\frac{R_2}{1/sC_1} = -sR_2C_1$

$$2) T = -\frac{x_r}{x_s} = A_d(s) \cdot \frac{sC_1}{1 + A_d(s)sC_1} \rightarrow \beta = \frac{1}{1 + sC_1R_2} \rightarrow W_{RC} = \frac{1}{C_1R_2} = 454 \text{ rad/s}$$

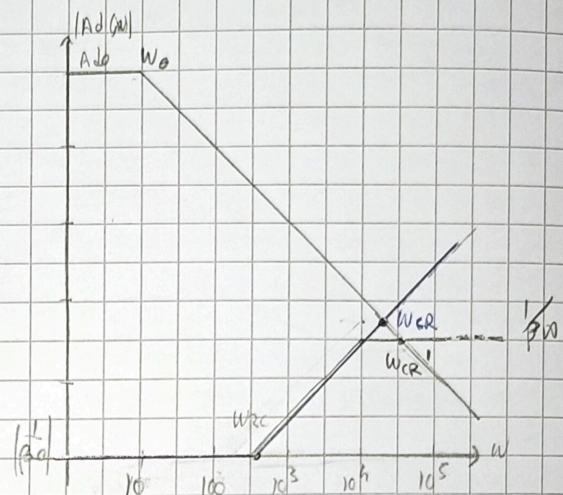
$$\beta_{f \rightarrow 0} = 1 \text{ B} \rightarrow \frac{1}{\beta_{f \rightarrow 0}} = 0 \text{ dB}$$

$$\beta_{f \rightarrow \infty} = 0 \rightarrow \frac{1}{\beta_{f \rightarrow \infty}} = \infty$$

$$\left\{ \begin{aligned} W_{CR} A_v &= A_{d0} \omega_0 \rightarrow W_{CR} = \sqrt{A_{d0} \omega_0 W_{RC}} \\ \frac{1}{\beta_0 W_{RC}} &= \frac{A_v}{W_{CR}} \end{aligned} \right. \rightarrow 21.31 \text{ Krad/s}$$

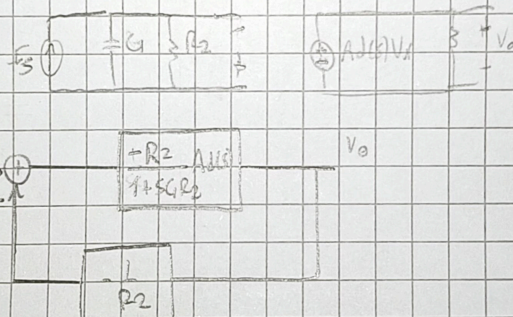
$$P_M = 180^\circ - \arg\left(\frac{W_{CR}}{\omega_0}\right) - \arg\left(\frac{W_{CR}}{W_{RC}}\right) = 1.25^\circ$$

large phase margin



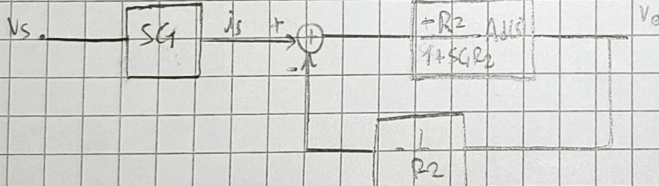
3) block diagram with feedback theory

current mixing, voltage sampling



$$\beta = -\frac{1}{R_2}$$

$$I_s = \frac{V_s}{Z_s} = sC_1 V_s$$



$$\frac{1}{sC_1} \parallel R_2 = \frac{R_2}{1 + sC_1R_2}$$

$$4) K_s(s) = SC_1$$

5) $P_M = 60^\circ$ with R_1 we are placing a zero $\rightarrow \beta_c = \frac{sC_1}{1 + R_1 + R_2} = \frac{1 + sC_1R_1}{1 + sC_1(R_1 + R_2)}$

$$\beta_{\infty} = \frac{R_1}{R_1 + R_2}$$

$$\frac{1}{\beta_{\infty}} = \frac{R_2}{R_1 + R_2}$$

to have a phase margin of 60°

$$P_M = 180^\circ - \arg\left(\frac{W_{CR}}{\omega_0}\right) - \arg\left(\frac{W_{CR}}{W_{RC}}\right) + \arg\left(\frac{W_{CR}'}{W_{Z_1}}\right) = 60^\circ$$

we can place the zero $W_{Z_1} = \frac{1}{C_1R_1}$ and we may consider that $W_{CR}' = \frac{1}{C_1(R_1 + R_2)}$

imposing $W_{CR}' = W_{RC}$

$$P_M = \arg\left(\frac{W_{CR}'}{W_{RC}}\right) + \arg\left(\frac{W_{CR}'}{W_{Z_1}}\right) = -30^\circ \rightarrow -\arg\left(\frac{W_{CR}'}{1/(C_1(R_1 + R_2))}\right) + \arg\left(\frac{W_{CR}'}{1/(C_1R_1)}\right) = -30^\circ$$

where the only unknown quantity is R_1

we can approximate $\frac{1}{C_1(R_1 + R_2)} \approx W_{RC} \rightarrow \arg\left(\frac{W_{CR}'}{1/(C_1R_1)}\right) = 58.78^\circ$

$$W_{Z_1} = 12.9 \text{ Krad/s}$$

$$W_{RC}' = 439 \text{ rad/s}$$

$$\beta_{\infty} = 0.034$$

$$W_{CR}' = \beta_{\infty} \cdot \omega_0 A_{d0} = 34 \text{ Krad/s}$$

$$R_1 = \frac{1}{W_{RC} C_1} = 774 \text{ }\Omega$$

(precise value would be $R_1 = 775 \text{ }\Omega$)

$$P_M = 70^\circ$$

$$R_1 = 600 \Omega$$

$$W_z = 16.6 \text{ Krad/s}$$

$$W_p = 442 \text{ rad/s}$$

$$W_{cr}' = 26.5 \text{ Krad/s}$$

$$\beta_w = 0.0265$$

$$\rightarrow PM = 59^\circ \quad (\checkmark)$$

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to calculate the precise 60° PM we need to consider that $W_{cr}' = \beta_w \cdot W_0 A_{db} = \frac{R_1}{R_1 + R_2} W_0 A_{db}$

$$\phi PM = 180^\circ - \arctg\left(\frac{W_{cr}'}{W_0}\right) - \arctg\left(\frac{W_{cr}'}{W_p}\right) + \arctg\left(\frac{W_{cr}'}{W_z}\right) = 60^\circ$$

$$W_z = \frac{1}{C R_1}$$

$$W_p = \frac{1}{C(R_1 + R_2)}$$

procedure for
exactly
 $60^\circ = PM$

$$= 120^\circ = \arctg\left(\frac{R_1}{R_1 + R_2} \cdot \frac{W_0 A_{db}}{W_0}\right) - \arctg\left(\frac{R_1}{R_1 + R_2} \cdot \frac{W_0 A_{db}}{\frac{1}{C(R_1 + R_2)}}\right) + \arctg\left(\frac{R_1}{R_1 + R_2} \cdot \frac{W_0 A_{db}}{\frac{1}{C R_1}}\right) = 0$$

$$= \frac{2}{3}\pi - \arctg\left(\frac{R_1 A_{db}}{R_1 + R_2}\right) - \arctg\left(R_1 C W_0 A_{db}\right) + \arctg\left(\frac{R_1^2 C W_0 A_{db}}{R_1 + R_2}\right) = 0$$

$$R_1 = \boxed{614 \Omega} \rightarrow W_z = 16.286 \text{ Krad/s}$$

$$W_p = 442.2 \text{ rad/s}$$

$$W_{cr}' = 27.151 \text{ Krad/s}$$

$$PM = 59.999^\circ \quad (\checkmark)$$